No Late-Time Step in Newton's Constant from Extra Dimensions, and the Screening Requirements of the Scale-Invariant Loophole

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A sharp, spatially uniform increase of the effective gravitational constant by $\sim 4\%$ at redshift $z \simeq 0.01$ (the "G step") has been proposed to resolve the Hubble tension, but no microphysical mechanism for it is known. We show that extra-dimensional compactifications cannot supply one. If G is controlled by a stabilized modulus, the transition between vacua shifts the four-dimensional vacuum energy by $|\Delta V| \simeq (c_K/k)\,\varepsilon\, m_\sigma^2 M_{\rm P}^2$, where $\varepsilon = \Delta G/G$, m_σ is the modulus mass and $c_K/k = \mathcal{O}(1)$. Fifth-force tests force $m_\sigma \gtrsim (50\,\mu{\rm m})^{-1}$, so $|\Delta V| \gtrsim 10^{58}\rho_\Lambda$. In an explicit six-dimensional Einstein–Maxwell flux compactification the coefficient is exactly $1/3$ at leading order in $1/n$. The bubble wall separating the two phases exceeds the Zel'dovich–Kobzarev–Okun bound by 31 orders of magnitude, and moving the Planck mass through a brane-induced term instead requires a quartic coupling below 10^{-117}. The only escape is a modulus that is light, as in a classically scale-invariant bulk. We show that in six-dimensional gauged chiral supergravity this fixes, with no free parameter, a Brans–Dicke scalar with $\omega = 1/2$, i.e. a fifth force of strength $\alpha = 1/4$ relative to gravity. Such a force lies within the range that screened-fifth-force analyses of the distance ladder find useful, but unscreened it violates the Cassini bound by four orders of magnitude, so the loophole converts the G step into an environmental-screening problem. We formalize what any screening mechanism (which we call a *dilaton brake coupling*) must achieve. These requirements are Solar-System suppression $\leq 4.6 \times 10^{-5}$, a switch-off across less than a decade of environmental potential with steepness $p \gtrsim 5$, an unscreened calibrator fraction $0.67 \lesssim f_u \lesssim 0.78$, and either a range below $\sim 0.3\,{\rm Mpc}$ or a selective screening of diffuse gas. Neither axio-chameleon nor chameleon screening meets these requirements. The allowed window brings H_0 to $\simeq 69$–$70\,{\rm km\,s^{-1}Mpc^{-1}}$, not to the Planck value, so the scale-invariant loophole can at most ease the Hubble tension.

I. INTRODUCTION

The local distance-ladder determination of the Hubble constant, $H_0 = 73.04 \pm 1.04\,{\rm km\,s^{-1}Mpc^{-1}}$ [1], disagrees with the value inferred from the cosmic microwave background in ΛCDM, $H_0 = 67.4 \pm 0.5$ [2], at the $\sim 5\sigma$ level. Among the proposed resolutions that act only on the local ladder, two are closely related. In the *G-step* model of Marra and Perivolaropoulos [3], the effective gravitational constant changes abruptly, $\Delta G/G \simeq 4\%$, at $z_t \simeq 0.01$, so that Type Ia supernovae in the Hubble flow and their Cepheid-calibrated hosts experience different gravity. This model remains phenomenologically debated [4, 5], but its mechanism is explicitly unknown. In the *screened fifth force* scenario of Desmond, Jain and Sakstein (DJS) [6], G does not change in time. Instead, Cepheids in unscreened, low-density hosts feel an additional scalar force that biases their periods relative to those in the screened anchor galaxies.

Extra dimensions are a natural place to look for a mechanism, since the four-dimensional Planck mass is set by the compactification volume, and Kaluza–Klein (KK) theories are routinely listed among candidate origins of a G step. This paper makes two contributions.

1. **A no-go result** (Sec. III). If the G step is produced by a transition between stabilized compactification vacua, whether through a flux jump, a brane-tension change or a volume transition, the four-dimensional vacuum energy shifts by $\gtrsim 10^{58}\rho_\Lambda$. We prove this in general (Proposition 1) and verify it exactly in six-dimensional Einstein–Maxwell theory. Two possible escapes, the bubble wall and a brane-induced Planck mass, fail independently.

2. **The scale-invariant loophole and its screening requirements** (Secs. IV–VI). The only escape is a light modulus. In a classically scale-invariant six-dimensional bulk, geometry fixes the resulting fifth-force strength to $\alpha = 1/4$. We compute its effect on the distance ladder and then derive, without assuming a specific mechanism, the quantitative conditions any environmental screening ("dilaton brake coupling") must satisfy.

We do not claim that the Hubble tension is resolved. The conclusion is sharper: extra dimensions cannot produce a G step, and the one extra-dimensional route to a local-G explanation requires a screening mechanism with properties no known mechanism has. A short section on gravitational-wave propagation (Sec. VII) and an appendix on normal-bundle ("twist") fields (Appendix B) record related results that do not affect the main argument.

Throughout, $M_{\rm P} = (8\pi G)^{-1/2} = 2.435 \times 10^{18}\,{\rm GeV}$, $\hbar = c = 1$, and $\rho_\Lambda \simeq 2.5 \times 10^{-11}\,{\rm eV}^4$.

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II. SETUP

We work in six dimensions with two compact transverse directions, the minimal setting for codimension-two branes and the one in which the relevant moduli dynamics is best understood [7–11]. The bulk action contains the Einstein–Hilbert term, a cosmological constant Λ_6, and a $U(1)$ gauge field with n units of magnetic flux through the transverse space:

$$S_6 = \int d^6X\sqrt{-G}\left[\tfrac{M_6^4}{2}\left(R_6 - 2\Lambda_6\right) - \tfrac{1}{4}F_{MN}F^{MN}\right] + S_{\rm br}. \tag{1}$$

The flux-stabilized solution is a "rugby ball": a sphere of radius ρ with conical deficits at the poles set by brane tensions, parametrized by $\lambda \le 1$, with metric

$$ds^2 = g_{\mu\nu}dx^\mu dx^\nu + \rho^2\left(d\theta^2 + \lambda^2\sin^2\theta\, d\varphi^2\right). \tag{2}$$

Matter is confined to a codimension-two brane at a pole and couples to the induced metric. The four-dimensional Planck mass receives a bulk contribution from the transverse area $A = 4\pi\lambda\rho^2$ and, if present, a brane-induced contribution F:

$$M_{\rm P}^2 = M_6^4\, A + F. \tag{3}$$

A fractional change $\varepsilon \equiv \Delta G/G$ felt by brane matter therefore requires either $\Delta A/A \simeq -\varepsilon$ (Sec. III) or $\Delta F \simeq -\varepsilon M_{\rm P}^2$ (Sec. III E). Sub-millimeter tests of the inverse-square law require $\rho \lesssim 50\,\mu$m [12, 13], and supernova SN1987A cooling requires $\rho \lesssim 1.2\,\mu$m for two extra dimensions of this shape [14] (converted from the torus bound $R < 0.66\,\mu$m at equal volume).

III. NO-GO THEOREM FOR AN EXTRA-DIMENSIONAL G STEP

A. Statement

Let $x \equiv \rho^{-2}$ and let the four-dimensional Einstein-frame potential of the volume modulus be a sum of terms generated by the bulk cosmological constant, curvature, flux and brane tensions,

$$V(x;\boldsymbol{p}) = \sum_j c_j(\boldsymbol{p})\, x^{k_j}, \tag{4}$$

where the microscopic parameters $\boldsymbol{p}$ (flux quanta, tensions, Λ_6) enter through the coefficients c_j. Brane matter sees $G \propto x$ at fixed M_6, so $\varepsilon = \Delta x/x_0$.

Proposition 1 (No G step from stabilized compactifications)**.** *Assume that (i) the volume modulus is stabilized at x_0 with $V(x_0) \simeq 0$ (to within ρ_Λ) before the transition, (ii) the transition changes a parameter that multiplies a single term x^k in Eq. (4), and (iii) no second, independent parameter changes simultaneously in a correlated way. Then the new vacuum has G shifted by ε and vacuum energy*

$$\begin{aligned}\Delta V_{\rm min} &= -\frac{\varepsilon}{k}\, x_0^2\, \frac{\partial^2 V}{\partial x^2}\bigg|_{x_0} + \mathcal{O}(\varepsilon^2) \\ &= -\frac{c_K}{k}\,\varepsilon\, m_\sigma^2 M_{\rm P}^2 + \mathcal{O}(\varepsilon^2),\end{aligned} \tag{5}$$

where m_σ is the mass of the canonically normalized modulus and $c_K = \mathcal{O}(1)$ is fixed by its kinetic term.

Proof. Let the parameter c multiply x^k, so that

$$\frac{\partial V}{\partial c} = x^k, \qquad \frac{\partial^2 V}{\partial x\,\partial c} = k\,x^{k-1}.$$

The minimum $x_0(c)$ is defined by $\partial V/\partial x = 0$. Differentiating this condition with respect to c,

$$\frac{\partial^2 V}{\partial x^2}\frac{dx_0}{dc} + \frac{\partial^2 V}{\partial x\,\partial c} = 0 \implies \frac{dx_0}{dc} = -\frac{k\,x_0^{k-1}}{\partial^2 V/\partial x^2}.$$

The value at the minimum, $V_{\rm min}(c) = V(x_0(c), c)$, changes by

$$\frac{dV_{\rm min}}{dc} = \underbrace{\frac{\partial V}{\partial x}}_{=0}\frac{dx_0}{dc} + \frac{\partial V}{\partial c} = x_0^k$$

(the envelope theorem). For a small change δc, $\varepsilon = \delta x_0/x_0 = -k\,x_0^{k-2}\,\delta c/(\partial^2 V/\partial x^2)$ and $\Delta V_{\rm min} = x_0^k\,\delta c$. Eliminating δc gives the first line of Eq. (5). For the second line, the canonical modulus ψ satisfies $x = x_* e^{-\psi/(bM_{\rm P})}$ with $b = \mathcal{O}(1)$, so $dx/d\psi = -x/(bM_{\rm P})$ and, by the chain rule,

$$m_\sigma^2 = \frac{d^2V}{d\psi^2}\bigg|_{x_0} = \left(\frac{dx}{d\psi}\right)^2\frac{\partial^2 V}{\partial x^2} + \frac{d^2x}{d\psi^2}\underbrace{\frac{\partial V}{\partial x}}_{=0} = \frac{x_0^2}{b^2M_{\rm P}^2}\,\frac{\partial^2 V}{\partial x^2},$$

which gives the second line with $c_K = b^2$. □

The proposition says the vacuum energy shift is *first order* in ε. The cancellation that makes $V(x_0)$ vanish is a cancellation between terms each of order $m_\sigma^2 M_{\rm P}^2$, and changing one of them by the fraction needed to move G by ε unbalances the sum by the same fraction.

B. Size of the effect

A modulus that controls G couples to brane matter with gravitational strength, so it is itself a Yukawa force of strength $\mathcal{O}(1)$. Torsion-balance tests then require its range to be below $\sim 50\,\mu$m [12, 13], i.e. $m_\sigma \gtrsim 4\times10^{-3}\,$eV. In a flux-stabilized rugby ball, $m_\sigma \sim \rho^{-1}$ and the same bound follows from $\rho \lesssim 50\,\mu$m. Equation (5) then gives

$$\frac{|\Delta V|}{\rho_\Lambda} \gtrsim 5\times10^{58}\,\frac{3\,c_K}{k}\left(\frac{\varepsilon}{0.04}\right)\left(\frac{m_\sigma}{4\,\text{meV}}\right)^2. \tag{6}$$

TABLE I. Vacuum-energy shift $|\Delta V| \simeq \varepsilon M_{\rm P}^2/\rho^2$ for a G step of $\varepsilon = 0.04$, in units of the observed dark-energy density. The exact Einstein–Maxwell coefficient (Sec. III C) multiplies these by 1/3.

ρ	$\lvert\Delta V\rvert/\rho_\Lambda$	Status of ρ
$50\,\mu$m	1.5×10^{59}	sub-mm gravity edge
$30\,\mu$m	4.1×10^{59}	allowed by sub-mm tests
$1.2\,\mu$m	2.5×10^{62}	SN1987A edge

TABLE II. Exact flux-sector transitions in the 6D Einstein–Maxwell model, Eq. (7), with Λ_6 tuned so sector n_0 is Minkowski. The ratio $|\Delta V|/(\varepsilon M_{\rm P}^2/\rho_0^2)$ approaches 1/3 as $n_0 \to \infty$.

n_0	new n	$\varepsilon = \Delta G/G$	$\Delta V/(M_{\rm P}^2/\rho_0^2)$	ratio
25	24	+0.125	−0.047	0.38
50	49	+0.061	−0.022	0.35
75	74	+0.041	−0.014	0.35
100	99	+0.030	−0.010	0.34
100	101	−0.030	+0.0096	0.32

Table I evaluates the estimate $|\Delta V| \sim \varepsilon M_{\rm P}^2/\rho^2$ (with c_K/k set to unity) for representative radii. Even at the largest radius allowed by laboratory tests, the shift exceeds the observed dark energy density by 59 orders of magnitude. At the SN1987A-allowed $\rho \simeq 1.2\,\mu$m it exceeds it by 62. Demanding $|\Delta V| \le \rho_\Lambda$ instead would require $\rho \gtrsim \sqrt{\varepsilon M_{\rm P}^2/\rho_\Lambda} \approx 4$–$6 \times 10^2$ Mpc, extra dimensions larger than the bubble whose interior is supposed to have different G.

C. Explicit check: 6D Einstein–Maxwell

In 6D Einstein–Maxwell theory on a sphere with n flux quanta, the four-dimensional Einstein-frame potential takes the form (units $M_6 = 1$; the gauge coupling and $\mathcal{O}(1)$ normalizations are absorbed into x and Λ_6) [7, 15]

$$V(x) = \frac{M_{\rm P}^4}{4\pi}\left(-x^2 + \Lambda_6\, x + n^2 x^3\right). \tag{7}$$

The three terms are the internal curvature, the bulk cosmological constant and the flux energy. Tuning $\Lambda_6 = 1/(4n_0^2)$ makes sector n_0 a Minkowski minimum at $x_0 = 1/(2n_0^2)$. Stepping to $n_0 \pm 1$ at fixed Λ_6 (Appendix A) gives, to leading order in $1/n_0$,

$$\varepsilon = \mp\frac{3}{n_0}, \qquad \frac{\Delta V}{M_{\rm P}^2/\rho_0^2} = \pm\frac{1}{n_0} = -\frac{\varepsilon}{3}, \tag{8}$$

which is Eq. (5) with $k = 3$. Table II shows the exact minima. A 4% step requires $n_0 \approx 75$, and the vacuum-energy jump is one third of the estimate in Table I.

D. The bubble wall

A spatially localized G step (a region of radius $R \sim 50$ Mpc around us) is bounded by a domain wall in the volume modulus. With field excursion $\Delta\psi \sim \varepsilon M_{\rm P}$ and mass $m_\sigma \sim \rho^{-1}$, the wall tension is

$$\sigma_w \sim m_\sigma (\Delta\psi)^2 \sim \varepsilon^2 \frac{M_{\rm P}^2}{\rho} \approx 6 \times 10^{49}\,\mathrm{eV}^3 \quad (\rho = 30\,\mu\mathrm{m}), \tag{9}$$

and its gravitational potential is $\Phi_w \sim \sigma_w R/M_{\rm P}^2 \sim 10^{25}$ at $R = 50$ Mpc. The Zel'dovich–Kobzarev–Okun bound on cosmological domain walls, $\sigma_w \lesssim (1\,\mathrm{MeV})^3$ [16], is violated by 31 orders of magnitude. A uniform global step avoids the wall but not Eq. (5).

E. The brane-induced Planck mass

Alternatively, G could change through the brane-induced term in Eq. (3), $F = \xi|\Phi|^2$ for a brane scalar with quartic self-coupling λ_Φ. Shifting F by $\varepsilon M_{\rm P}^2$ moves $|\Phi|^2$ by $\varepsilon M_{\rm P}^2/\xi$ and costs brane vacuum energy $\sim \lambda_\Phi(\varepsilon M_{\rm P}^2/\xi)^2$. Requiring this to be below ρ_Λ gives

$$\frac{\lambda_\Phi}{\xi^2} \lesssim \frac{\rho_\Lambda}{\varepsilon^2 M_{\rm P}^4} \approx 5 \times 10^{-118}, \tag{10}$$

a second cosmological-constant problem. Energy that is not cancelled in the bulk reappears as four-dimensional curvature.

F. Scope and the escape clause

Proposition 1 covers flux jumps, tension changes and volume transitions between stabilized vacua, including the football-shaped compactifications of Refs. [7, 8]. Garriga and Porrati showed that such compactifications do not self-tune [9], and Brown and Dahlen analyzed flux-changing transitions in the same landscape [15]. To our knowledge the consequence for a late-time G step has not been stated. It evades only three loopholes:

1. *Correlated tuning:* two independent parameters (say flux and tension) jump together so that $\sum_j x_0^{k_j}\delta c_j = 0$. This requires a coincidence at the level of $\rho_\Lambda/(\varepsilon m_\sigma^2 M_{\rm P}^2) \lesssim 10^{-58}$, with flux quantized.

2. $\mathcal{O}(1)$ *prefactors:* these cannot close a gap of 58 orders of magnitude.

3. *A light modulus:* if m_σ is not bounded below because the bulk has a flat direction, Eq. (6) no longer applies. But then the modulus mediates a long-range, gravitational-strength fifth force, and G felt by matter depends on environment. A homogeneous step cannot occur: a field with $m_\sigma \gg H_0$ tracks the slowly varying mean density, and one

with $m_\sigma \lesssim H_0$ evolves on a Hubble time, not within $\Delta z \ll 0.01$. This is the subject of Secs. IV–VI.

Proposition 1 is therefore a dichotomy. A heavy modulus gives a vacuum-energy catastrophe. A light modulus turns the G step into a screened-fifth-force problem.

IV. THE SCALE-INVARIANT LOOPHOLE: $\alpha = 1/4$ FROM GEOMETRY

Six-dimensional gauged chiral (Salam–Sezgin) supergravity [17], the basis of the supersymmetric large-extra-dimensions (SLED) program [10, 18], has a classical scaling symmetry $g_{MN} \to \zeta g_{MN}$, $e^{\phi} \to \zeta^{-1} e^{\phi}$. A change of brane tension can then be absorbed by sliding along the flat direction $e^{\phi}\rho^2 = \text{const}$ while the brane stays flat. This is precisely the light-modulus escape.

Write the 6D metric as $g_{\mu\nu}(x)dx^\mu dx^\nu + e^{2\sigma(x)}\rho_0^2\, d\Omega_2^2$ and restrict to the flat direction, $\phi = -2\sigma + \text{const}$. With $\Phi \equiv e^{2\sigma}$, the four-dimensional action in the frame brane matter sees (the induced metric) is of Brans–Dicke form,

$$S = \frac{M_0^2}{2}\int d^4x\sqrt{-g}\left[\Phi R - \frac{\omega}{\Phi}(\partial\Phi)^2\right],$$
$$\omega = \underbrace{-\tfrac{d-1}{d}}_{\text{volume},\, d=2} + \underbrace{1}_{\text{dilaton}} = \tfrac{1}{2}. \tag{11}$$

The volume contribution is the standard KK result ($\omega = 0$ for one extra dimension, $-1/2$ for two). The dilaton contributes $+1$ because its 6D kinetic term is normalized like the curvature term and $(\partial\phi)^2 = (\partial\Phi)^2/\Phi^2$. Since brane matter couples only to the induced metric, the fifth-force strength relative to gravity, the scalar–matter coupling and the post-Newtonian parameter follow directly:

$$\alpha \equiv \frac{F_{\text{scalar}}}{F_{\text{grav}}} = \frac{1}{2\omega+3} = \frac{1}{4}, \tag{12}$$
$$\beta = \sqrt{\alpha/2} \simeq 0.35, \qquad \gamma_{\text{PPN}} = \frac{1+\omega}{2+\omega} = 0.6.$$

The strength is fixed by the codimension and the scaling symmetry. It is not a fitted parameter.

Range. Loops lift the flat direction and give the modulus a mass $m \sim 1/(\rho^2 M_{\text{P}})$ [18]. Table III lists the resulting range together with the 6D scale M_6 implied by Eq. (3).

Unscreened, it is excluded. Equation (12) gives $|\gamma_{\text{PPN}} - 1| = 0.4$, while Cassini requires $|\gamma_{\text{PPN}} - 1| < 2.3 \times 10^{-5}$ [19]. Since $\gamma - 1 = -2\alpha_{\text{eff}}/(1+\alpha_{\text{eff}})$, the effective coupling of the Sun must be suppressed by

$$\frac{\alpha_{\text{eff},\odot}}{\alpha} \le 4.6 \times 10^{-5}. \tag{13}$$

Old neutron-star heating gives $\rho \lesssim 0.09\,\mu$m [14] unless KK gravitons decay mostly into the bulk. Either way, the range remains far larger than a stellar radius, which is all the Cepheid effect below requires.

TABLE III. Dilaton range $1/m$ with $m \sim 1/(\rho^2 M_{\text{P}})$ (uncertain by an $\mathcal{O}(1)$ factor) and the 6D Planck scale implied by Eq. (3) with $\lambda = 1$.

ρ	range	M_6	status
$10\,\mu$m	40 Mpc	3.7 TeV	excluded (SN1987A, morphology)
$4.5\,\mu$m	8 Mpc	5.5 TeV	excluded (SN1987A, morphology)
$1.2\,\mu$m	0.6 Mpc	11 TeV	SN1987A edge
$0.87\,\mu$m	0.3 Mpc	13 TeV	below morphology band
$0.09\,\mu$m	3 kpc	39 TeV	neutron-star-heating edge

Caveats. The value $\omega = 1/2$ is classical. Brane couplings such as a dilaton-dependent tension $T(\phi)$, localized flux [20], warping near the conical poles and loop corrections can shift it. A first-principles derivation of β from the brane action is left to future work.

V. EFFECT ON THE DISTANCE LADDER

Following DJS [6], suppose Cepheid envelopes in a fraction f_u of supernova host galaxies feel the full $\alpha = 1/4$ force, while the anchor Cepheids (Milky Way, NGC 4258) are screened. A stronger effective G shortens the pulsation period, $P \propto (G\bar{\rho})^{-1/2} \propto (1+\alpha)^{-1/2}$. With the near-infrared Wesenheit period–luminosity slope $\log L \simeq 1.3 \log P$, each unscreened Cepheid's distance is underestimated by

$$\Delta \log_{10} d = -0.325 \log_{10}(1+\alpha),$$
$$\frac{\Delta d}{d} = -7.0\%, \qquad \Delta\mu = 0.157\,\text{mag}. \tag{14}$$

Underestimated calibrator distances make the supernovae appear too faint, so H_0 is biased high. To linear order the corrected value is $H_0 = 73.04\,(1 - 0.070 f_u)$. We neglect the smaller change in luminosity at fixed period, which DJS include.

The fifth force also acts on the tip of the red giant branch (TRGB), but we conservatively take TRGB distances to be unaffected. JWST finds Cepheid and TRGB distance moduli to the same hosts in agreement, $\Delta\mu_{\text{C-T}} = 0.01 \pm 0.04\,(\text{stat}) \pm 0.04\,(\text{sys})$ [21], while the fifth force predicts $\Delta\mu_{\text{C-T}} = 0.157 f_u$. Table IV shows the trade-off. If TRGB luminosities are also shifted [22], part of the Cepheid–TRGB offset cancels, but the TRGB-calibrated H_0 is lowered too.

Requiring both tensions to be below 2σ selects a narrow window,

$$0.67 \lesssim f_u \lesssim 0.78, \qquad H_0 \simeq 69\text{–}70\,\text{km}\,\text{s}^{-1}\text{Mpc}^{-1}. \tag{15}$$

The full DJS analysis with realistic screening maps finds that many models reduce the tension below 3σ, but reduction below 2σ is unlikely [6]. Our linear estimate should be read in that light. It shows that the geometrically fixed $\alpha = 1/4$ has the right order of magnitude, not that the tension is resolved.

TABLE IV. Linearized effect of an $\alpha = 1/4$ force on a fraction f_u of unscreened calibrator Cepheids, starting from $H_0 = 73.04 \pm 1.04$. "Planck" is the tension with 67.4 ± 0.5; "C–T" is the tension with the JWST Cepheid–TRGB agreement, assuming TRGB is unaffected.

f_u	H_0	Planck	$\Delta\mu_{\mathrm{C-T}}$ (mag)	C–T
0.25	71.8	3.8σ	0.04	0.5σ
0.50	70.5	2.8σ	0.08	1.2σ
0.75	69.2	1.6σ	0.12	1.9σ
1.00	67.9	0.5σ	0.16	2.6σ

VI. REQUIREMENTS ON A DILATON BRAKE COUPLING

Section IV fixes the strength of the force, and Sec. V shows what it would do if suitably screened. Whether a screening mechanism exists is a separate question. Rather than posit one, we define the class of mechanisms required and derive its constraints.

Definition 1. A *dilaton brake coupling* is an environment-dependent suppression of the scalar charge of a body i,

$$\beta_{\mathrm{eff}}^{(i)} = \beta\,\mathcal{B}_i, \qquad 0 \le \mathcal{B}_i \le 1, \tag{16}$$

so that the force between two bodies has strength $\alpha\,\mathcal{B}_i\mathcal{B}_j$ (and $\alpha\,\mathcal{B}_i$ in one-sided tests such as light deflection), where $\mathcal{B}_i$ may depend on the body's own potential Φ_{self}, its environmental potential Φ_{env} and its local density, and must reduce to $\mathcal{B} = 1$ in the unscreened limit so that the geometric prediction $\alpha = 1/4$ is preserved.

The data impose the following requirements.

R1 (Solar System). By Eq. (13), $\mathcal{B}_\odot \lesssim 4.6 \times 10^{-5}$ for the one-sided light-deflection and time-delay test.

R2 (Anchors screened). Cepheids in the Milky Way and NGC 4258 must have $\mathcal{B} \ll 1$. Otherwise the anchor and host biases cancel.

R3 (Hosts unscreened). Envelopes of Cepheids in a fraction f_u of supernova hosts must have $\mathcal{B} \simeq 1$, with f_u in the window of Eq. (15).

R4 (Self-screening cannot do it). The Sun's surface potential (2×10^{-6}) is only ten times a Cepheid's ($\sim 2 \times 10^{-7}$ for $R \approx 50R_\odot$). A brake depending on Φ_{self} alone cannot reach $\mathcal{B}_\odot < 5 \times 10^{-5}$ while keeping $\mathcal{B}_{\mathrm{Ceph}} \simeq 1$ unless it switches more steeply than any thin-shell mechanism. The suppression must instead come from the environment.

R5 (Steepness). Model the environmental dependence as $\mathcal{B} = [1 + (\Phi_{\mathrm{env}}/\Phi_b)^p]^{-1}$ and let $\kappa \equiv \Phi_{\mathrm{env,MW}}/\Phi_{\mathrm{env,host}}$. Satisfying R1 ($\mathcal{B} \le 4.6 \times 10^{-5}$ in the Milky Way environment) together with R3 ($\mathcal{B} \ge 0.9$ in hosts) requires

$$p \;\ge\; \frac{\ln(2.0 \times 10^5)}{\ln \kappa}, \tag{17}$$

i.e. $p \ge \{11, 5.3, 3.6, 2.7\}$ for $\kappa = \{3, 10, 30, 100\}$. The brake must switch off over at most one to two decades of environmental potential.

R6 (Galaxy morphology). Low-density galaxies whose environment leaves Cepheid envelopes unscreened also host diffuse gas. If the gas feels the force while stars partly do not, gas–star offsets and warps result. Morphology limits the force strength to $\alpha\mathcal{B}_{\mathrm{gas}}^2 \lesssim 0.05$ at a range of 1.2 Mpc, tightening to 3×10^{-5} at 8 Mpc [23, 24], with sensitivity over roughly 0.3–8 Mpc. For $\alpha = 1/4$ either

(a) the range is $\lesssim 0.3$ Mpc, which requires $\rho \lesssim 0.9\,\mu$m (Table III) and is consistent with SN1987A, or

(b) the brake screens diffuse gas ($\mathcal{B}_{\mathrm{gas}}^2 \lesssim 0.2$) while leaving stellar envelopes in the same galaxies unscreened, the reverse of the ordering produced by thin-shell mechanisms.

R7 (Range consistency). In chameleon (thin-shell) screening a Cepheid envelope is unscreened only if the threshold $\chi_c \gtrsim 2 \times 10^{-7}$. With the Desmond–Ferreira relation $\chi_c \simeq 1.5 \times 10^{-8}(\lambda_C/0.32\,\mathrm{Mpc})^2$ [23], this requires $\lambda_C \gtrsim 1.2$ Mpc, above the $\lesssim 0.6$ Mpc allowed by SN1987A (up to the $\mathcal{O}(1)$ uncertainty in the range). A chameleon-type realization therefore conflicts with branch (a) of R6.

A. Known mechanisms

Axio-chameleon screening. This is the native mechanism for the axio-dilaton of string-inspired and SLED-like constructions [25, 26]. Here the dilaton charge of a source of radius R is reduced by the axion gradient in a surface layer of width ℓ, $\mathfrak{g}_{\mathrm{eff}} = \mathfrak{g}/[1 + \frac{1}{4}(R/\ell)\widehat{W}_1^2(\mathfrak{a}_+ - \mathfrak{a}_-)^2]$ [25]. It fails R3 and R5 for two reasons. Larger bodies are *more* screened, so a Cepheid ($R \approx 50R_\odot$) is screened about 50 times more strongly than the Sun, which R1 already requires to be screened. And screening requires the external axion value to differ from the internal one, so denser surroundings screen *less*, the opposite of R2 and R3.

Chameleon screening. This leaves Cepheid envelopes exposed only for $\lambda_C \gtrsim 1.2$ Mpc (R7). There galaxy morphology limits the force to $\alpha \lesssim 0.05$, a factor 4–5 below $\alpha = 1/4$. A chameleon of strength 0.05 could pass, but then the coupling is no longer the one the geometry predicts.

Density-threshold variants. These unscreen only gas below a critical density. They leave the dense Cepheid core screened, so only the small envelope mass feels the force, far too little to shift periods by percent levels.

B. Summary of the map

Table V collects the requirements. Branch (a) of R6 is the most economical. A sub-micron rugby ball ($\rho \lesssim 0.9\,\mu$m, $M_6 \gtrsim 13$ TeV) puts the dilaton range

below the morphology band, and what remains is an environmental brake satisfying R1–R5. No known mechanism provides this, but the target is now precise. It is an environment-dependent (not self-potential-dependent) suppression with steepness $p \gtrsim 5$ across a factor ~ 10 in environmental potential, acting on a gravitational-strength scalar of range between a stellar radius and $\sim 0.3\,\mathrm{Mpc}$.

VII. GRAVITATIONAL-WAVE SECTOR

The same geometry implies a gravity-side observable. This section is not needed for the results above, and the underlying effect, gravitational waves taking a bulk "shortcut", is known [27–30]. Light follows the induced brane metric $g_{\mu\nu}^{\rm light} = \eta_{\mu\nu} + \partial_\mu y^I \partial_\nu y^I$, a disformal metric with the brane-bending modes (branons [31, 32]) as the scalar. The graviton zero mode is uniform over the compact space and follows the background metric up to its brane-localized fraction $f_{\rm loc} = F/M_{\rm P}^2$. Along a line of sight,

$$\frac{\Delta t}{t} \simeq \tfrac{1}{2}(1 - f_{\rm loc})\langle \dot{y}^2 + (\partial_\parallel y)^2 \rangle \;\geq\; 0, \qquad (18)$$

so gravity never trails light. The new element is the $(1 - f_{\rm loc})$ suppression by brane-induced gravity, which we have not found in the literature. GW170817 [33] bounds $\Delta t/t \lesssim 4 \times 10^{-16}$. If the displacement mode sits in an unstable flux sector, so that its amplitude is set by $|m^2_{\rm min}(n)|$, the bound requires that mode to be $\gtrsim 10^4$ times lighter than the brane-tension scale. Determining the spectrum $m^2_{\rm min}(n)$ is left for future work.

VIII. CONSISTENCY AND LIMITATIONS

What the main results depend on. Proposition 1 uses only Eq. (3), the structure (4) of the moduli potential and fifth-force bounds. The value $\alpha = 1/4$ uses only the classical scale symmetry and minimal coupling to the induced metric. Neither depends on the brane rigidity or brane-induced gravity terms discussed next.

Ghosts. Two ingredients of the broader framework carry known ghost risks and are excluded from the main results. (i) An extrinsic-rigidity term $\alpha_r K_I K^I$ contains $(\Box y)^2$ for the branons and generically introduces a higher-derivative mode at mass $\sim\sqrt{T/\alpha_r}$, as for the rigid string. It must be treated as an effective-field-theory correction valid below that scale. (ii) Brane-induced gravity with more than one extra dimension has a known ghost in infinite-volume setups [34]. Whether compactness of the rugby ball removes it should be checked before relying on $f_{\rm loc}$ (Sec. VII).

Diffeomorphism invariance. All constructions here are fully covariant in six dimensions. The extrinsic quantities are contracted over the normal index and are invariant under rotations of the normal frame. No preferred foliation is introduced.

Open issues. The $\mathcal{O}(1)$ coefficient c_K/k in Proposition 1 has been computed exactly only for the Einstein–Maxwell sphere; conical defects modify it by $\mathcal{O}(1)$ factors, which cannot affect the conclusion. The value $\omega = 1/2$ relies on the classical SLED flat direction, which is subject to the self-tuning critiques of Refs. [9, 10] and may shift once brane-localized couplings are included; Secs. IV–VI should be read with that caveat.

IX. CONCLUSIONS

A sharp, late-time step in G cannot come from extra dimensions with stabilized moduli. The vacuum energy shifts at first order in $\Delta G/G$, by $\gtrsim 10^{58}\rho_\Lambda$, and the alternatives of a localized bubble or a brane-induced Planck mass fail independently. The only escape is a light modulus. In a scale-invariant six-dimensional bulk, that modulus produces a fifth force of fixed strength $\alpha = 1/4$, of the right size for screened-fifth-force corrections to the distance ladder. Its viability rests entirely on a screening mechanism. We have specified that mechanism's requirements quantitatively: environmental rather than self-screening, a steep switch ($p \gtrsim 5$ per decade of environmental potential), an unscreened calibrator fraction 0.67–0.78, and either sub-0.3 Mpc range or selective gas screening. Even in the allowed window, H_0 falls to $\simeq 69$–70, easing but not removing the tension. Known screening mechanisms fail these requirements. Constructing one that satisfies them, or proving none exists, is the open problem this work leaves.

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This line of inquiry began as a heuristic: while reverse-engineering a zero-knowledge-proof program concept, the author pictured a hidden bulk whose curvature parameters are constrained by what an observer on the brane can verify. The physics presented here does not depend on that analogy. The author used an AI assistant (Claude, Anthropic) for checking calculations, literature searching and editing. All results and any errors are the author's responsibility.

Appendix A: Einstein–Maxwell flux transitions

For Eq. (7), write $v(x; n) = -x^2 + \Lambda_6 x + n^2 x^3$, with derivatives

$$\frac{\partial v}{\partial x} = -2x + \Lambda_6 + 3n^2x^2, \qquad \frac{\partial^2 v}{\partial x^2} = -2 + 6n^2 x. \quad \text{(A1)}$$

TABLE V. Requirements on any dilaton brake coupling acting on the $\alpha = 1/4$ scale-invariant-bulk scalar, and the status of known screening mechanisms. A check mark means the requirement can be met; a cross means it fails.

	Requirement	Quantitative target	Axio-chameleon	Chameleon
R1	Solar System screened	$\mathcal{B}_\odot \leq 4.6 \times 10^{-5}$	✓	✓
R2	Anchors screened	$\mathcal{B}_{\rm MW}, \mathcal{B}_{\rm N4258} \ll 1$	× (dense env. screens less)	✓
R3	Host envelopes unscreened	$\mathcal{B} \simeq 1$, $0.67 \lesssim f_u \lesssim 0.78$	× (large R screened)	only if $\lambda_C \gtrsim 1.2\,$Mpc
R4	Environmental, not self, screening	$\Phi_\odot/\Phi_{\rm Ceph} \approx 10$ insufficient	×	✓
R5	Steep switch	$p \geq \ln(2 \times 10^5)/\ln\kappa$	—	✓(thin shell)
R6	Morphology	range $\lesssim 0.3\,$Mpc, or gas screened	—	× (factor 4–5)
R7	Range consistent with SN1987A	$\lambda \lesssim 0.6\,$Mpc	—	× (needs $\geq 1.2\,$Mpc)

Requiring a Minkowski minimum in sector n_0, $v = \partial v/\partial x = 0$, gives

$$x_0 = \frac{1}{2n_0^2}, \qquad \Lambda_6 = \frac{1}{4n_0^2}, \qquad \left.\frac{\partial^2 v}{\partial x^2}\right|_{x_0} = 1 > 0. \quad \text{(A2)}$$

For $n \to n_0 + \delta n$ at fixed Λ_6, setting $\partial v/\partial x = 0$ gives $3n^2x^2 - 2x + \Lambda_6 = 0$, whose minimum ($\partial^2 v/\partial x^2 > 0$) is the larger root,

$$x_1 = \frac{1 + \sqrt{1 - 3n^2\Lambda_6}}{3n^2}. \quad \text{(A3)}$$

To first order, with $\delta(n^2) = 2n_0\,\delta n$ and $\partial^2 v/\partial x\,\partial(n^2) = 3x^2$,

$$\delta x = -\frac{\partial^2 v/\partial x\,\partial(n^2)}{\partial^2 v/\partial x^2}\,\delta(n^2) = -6n_0x_0^2\,\delta n, \quad \text{(A4)}$$

$$\varepsilon = \frac{\delta x}{x_0} = -\frac{3\,\delta n}{n_0}, \quad \text{(A5)}$$

$$\delta v_{\rm min} = \left.\frac{\partial v}{\partial(n^2)}\right|_{x_0}\delta(n^2) = x_0^3\,\delta(n^2) = 2n_0x_0^3\,\delta n. \quad \text{(A6)}$$

With $M_{\rm P}^2 = 4\pi\rho_0^2$ ($M_6 = 1$),

$$\frac{\Delta V}{M_{\rm P}^2/\rho_0^2} = \frac{\delta v_{\rm min}}{x_0^2} = \frac{\delta n}{n_0} = -\frac{\varepsilon}{3}. \quad \text{(A7)}$$

Table II uses the exact root x_1.

Appendix B: The twist field and photon mixing

For completeness we record a null result. At codimension two the rotation of the two brane normals defines the normal-bundle connection $B_\mu = n_2 \cdot \nabla_\mu n_1$. Its field strength is fixed by the Ricci equation of the embedding, and at a pole of the rugby ball it coincides (up to an $\mathcal{O}(1)$ factor of λ) with the KK gauge field W_μ of the transverse rotation [35]. It is not an independent field. Its gauge coupling is gravitational, $g_W \sim 1/(\lambda\rho M_{\rm P})$. It mixes with the bulk gauge zero mode through the vortex winding and acquires a mass for $n \neq 0$, consistent with the gauge spectra of Ref. [36]. Kinetic mixing with the photon can arise only from loops of brane fields carrying both electric charge and transverse spin, $\epsilon \sim e g_W/(16\pi^2) \sim e/(16\pi^2\lambda\rho M_{\rm P})$, so $\epsilon \sim 10^{-30}$ at $\rho = 10\,\mu$m. Photon–twist oscillations then violate the distance-duality relation $d_L = (1+z)^2 d_A$ at order $\epsilon^2 \sim 10^{-60}$, unobservable in supernova or BAO data. Light-based distance tests are therefore insensitive to this sector unless some term raises the coupling far above gravitational strength, and no such term appears in the action. The observable signatures of the framework lie on the gravity side.

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